Fabric defect detection: a Roboflow and PyTorch comparison

Single-label classification of fabric images into Defect free, Defected, and Waste. Both models are ResNet-18, trained on the same 70/20/10 split.

1. Dataset

The images were labeled in Roboflow and frozen as dataset version 1 (Poorva Fabric defect detection, IDM 2), exported in folder format on 29 October 2026. The export contains 4,995 source images: 3,499 train, 999 validation, and 497 test, which is the requested 70/20/10 split. Every image was auto-oriented and stretched to 224x224. Two augmentations were applied to the training images only: a horizontal flip with probability 50%, and a brightness change between -15% and +15%. Roboflow kept two copies of each training image, so the training folder has 6,998 files.

The classes are uneven. The training folder holds 6,304 Defected, 670 Defect free, and 24 Waste images. The test folder holds 449 Defected, 47 Defect free, and 1 Waste image. A model that almost always predicts Defected will score well on overall accuracy. Per-class recall is the number that shows whether Defect free and Waste are actually recognized.

2. The two models

The Roboflow model is sanya-mohanty/poorva-fabric-defect-detection-idm-2-1-resnet18-t1. It is a ResNet-18 started from the ImageNet checkpoint, trained with Roboflow's default hyperparameters. Inference in this benchmark calls the hosted API at serverless.roboflow.com. On the test split, Roboflow's evaluation reports precision 94.5%, recall 91.7%, and F1 93.0% at an operating threshold of 61% confidence.

The PyTorch model is also a ResNet-18, loaded from ImageNet weights with the final layer replaced by three outputs, then fine-tuned on the exported train and validation folders. Training used AdamW with learning rate 0.0003, weight decay 0.0001, batch size 32, and a cosine learning-rate schedule for 15 epochs. The training loader applied the same two augmentations, random horizontal flip and brightness jitter of 15%. Validation and test images were only resized and normalized. The run used the Apple GPU (MPS), took 19.2 minutes (1154 seconds), and reached its best validation accuracy of 97.0% at epoch 14.

Both models are scored on one shared list of 50 test images, saved in eval_images.txt so the comparison cannot drift. The list has 25 Defect free, 24 Defected, and 1 Waste image. It contains every Waste test image and about half of the Defect free test images, so it is a harder sample than the full test set. predictions.csv holds the Roboflow output for these 50 images: file name, predicted label, and confidence.

3. Accuracy, training time, and latency

Measure	Roboflow ResNet-18	PyTorch ResNet-18
Test precision	94.5%	92.7% macro
Test recall	91.7%	90.2% macro
Test F1	93.0%	91.4% macro
Accuracy on full test set (497)	96.4% (479/497)	95.8% (476/497)
Accuracy on the same 50 images	82% (41/50)	78% (39/50)
Training time	20 min	19.2 min
Mean latency on the 50	382 ms	11.3 ms
Median latency on the 50	365 ms	11.2 ms

The held-out test split is 497 images, the full 10% split, not a 500-image round number. PyTorch predicts a class for every one of them and is correct on 476, which is 95.8% accuracy. That score is mostly the Defected class: recall is 98.2% (441 of 449). Defect free recall is 72.3% (34 of 47). The single Waste image is correct, and one image is not evidence. Roboflow's confusion matrix on this same 497-image split has 36 Defect free, 442 Defected, and 1 Waste on the diagonal, which is 479 correct and 96.4% accuracy. Ten Defect free images are called Defected, six Defected images are called Defect free, and two images fall below the confidence threshold, one from each of those classes. At the 61% threshold Roboflow reports precision 94.5%, recall 91.7%, and F1 93.0%. PyTorch precision, recall, and F1 in the table are macro averages, so each class counts equally.

On the shared 50 images, Roboflow is correct on 41 and PyTorch on 39. The four-point gap is two images. Both percentages sit well below the full-test accuracy because this sample is half Defect free, and Defect free is the class both models miss.

Latency is where the two setups differ. Roboflow's mean of 382 ms per image (median 365 ms, 95th percentile 515 ms) includes sending the file to the hosted API and waiting for the response. PyTorch's mean of 11.3 ms is measured on the same machine and includes loading the image and preprocessing; the forward pass alone averages 8.8 ms. The hosted number is the cost of calling the API. The local number is the cost of running the network beside the camera.

The Roboflow ResNet-18 training job took 20 minutes. PyTorch training took 19.2 minutes for 15 epochs, about 77 seconds per epoch, on a laptop GPU. The hosted job used Roboflow's default recipe. The local job used the learning rate, batch size, and epoch count in section 2.

4. Failure analysis

Of the 50 shared images, 11 are wrong in at least one model, and every one of them is a Defect free image predicted as Defected. Nine are wrong in both models, most of them with confidence above 0.97. Two are wrong only in PyTorch. On defect-free-137, Roboflow scores Defect free at 0.616 and PyTorch scores Defected at 0.899. On defect-free-54, Roboflow scores Defect free at 0.638 and PyTorch scores Defected at 0.613. Those two Roboflow decisions sit just above the 61% threshold, so they are shallow wins.

The full PyTorch test set shows the same boundary. It makes 21 errors: 13 Defect free images called Defected, and 8 Defected images called Defect free. Waste is never confused with either class. The PyTorch confusion matrix is:

True \ predicted	Defect free	Defected	Waste
Defect free	34	13	0
Defected	8	441	0
Waste	0	0	1

The missed images are hard to tell apart by eye once they have been stretched to 224x224 and compressed. Many of the files are only 2 to 5 KB. A plain measure of texture, the standard deviation of the grayscale pixels, lines up with the mistakes. A typical Defect free test image has a standard deviation near 3. The Defected images called Defect free are the smoothest Defected images, down to 2.2. The Defect free images called Defected include the most textured Defect free images, with a standard deviation near 16, close to the Defected median of 11.8. Both networks are separating swatches by how much the weave varies, and on these edge cases that variation does not match the label. Because the wrong answer often comes with confidence above 0.99, raising the threshold does not remove them. The Waste test image is a fold rather than a stain, and both models label it Waste. With 12 Waste images in the original training split and one in the test split, that single correct call is not a result to lean on.

5. Conclusion

The two ResNet-18 trainings land in the same place on accuracy. Roboflow's published test F1 is 93.0%, and the PyTorch macro F1 is 91.4%. On the 50 shared images Roboflow is two images ahead, 82% against 78%. The local model answers in about 11 ms, against about 382 ms for the hosted API. Roboflow trained in 20 minutes, and the local run took 19.2 minutes on a laptop. The errors are concentrated on one boundary: smooth Defected swatches and textured Defect free swatches. More images in Defect free and Waste, and a pass over the labels where those two classes look alike, would change the result more than a larger network would.